

MATH 220/199 Week 8 Day 1

Reminders

1. Office hours are held in Davenport 330 at 3pm on Thursdays.
2. Do the Merit Quiz on the Merit Canvas page.

We can do Infinitely Better than Linearization

Linearization works because we have a reference point and understand the nearby behavior of a graph near the point of interest - What if we also knew about the behavior of the behavior of the graph? Let's estimate $(2.01)^3$

1. First, using linearization, find an estimate for $(2.01)^3$. Call your linearization equation $L_1(x)$.

2. Define $L_2(x) = L_1(x) + \frac{f''(a)}{2!}(x-a)^2 = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2$. Use $L_2(x)$ to estimate $(2.01)^3$. Is this estimate better?

3. Define $L_3(x) = L_2(x) + \frac{f'''(a)}{3!}(x-a)^3$ and re-estimate $(2.01)^3$

4. In general, we can construct these equations which give us better and better estimates using the recursive formula $L_n(x) = L_{n-1}(x) + \frac{f^{(n)}(a)}{n!}(x-a)^n$ - in Calc 2 you will call this a "Taylor Series". If you have time, compute $L_4(2.01)$ and see how good this estimate is.

Plausible Exam Word Problem

Suppose that a particle oscillates and has position function given by $p(t) = \sin(t) - \cos(t)$

1. Over the interval $[0, 3\pi]$, determine all the times t when the particle is not moving.
2. Over the interval $[0, \pi]$, determine the intervals where the particle is speeding up, and where it is slowing down.
3. Over the interval $[0, \pi]$, determine when the particle has the greatest speed.
4. The change in acceleration over time is called "jerk". If $v(t)$ is the velocity equation, find $j(t)$, the jerk equation in terms of $v(t)$.